

# Towards Meta-Cognitive Knowledge Editing for Multimodal LLMs

Zhaoyu Fan\*  
Zhejiang University  
Hangzhou, China  
zyfan@zju.edu.cn

Bosheng Qin  
Zhejiang University  
Hangzhou, China  
bsqin@zju.edu.cn

Wenqiao Zhang  
Zhejiang University  
Hangzhou, China  
wenqiaozhang@zju.edu.cn

Kaihang Pan\*<sup>†</sup>  
Zhejiang University  
Hangzhou, China  
kaihangpan@zju.edu.cn

Juncheng Li  
Zhejiang University  
Hangzhou, China  
junchengli@zju.edu.cn

Siliang Tang<sup>‡</sup>  
Zhejiang University  
Hangzhou, China  
siliang@zju.edu.cn

Yueting Zhuang  
Zhejiang University  
Hangzhou, China  
yzhuang@zju.edu.cn

Mingze Zhou  
Zhejiang University  
Hangzhou, China  
mingze@zju.edu.cn

Shengyu Zhang  
Zhejiang University  
Hangzhou, China  
sy\_zhang@zju.edu.cn

Fei Wu  
Zhejiang University  
Hangzhou, China  
wufei@zju.edu.cn

## Abstract

Knowledge editing enables multimodal large language models (MLLMs) to efficiently update outdated or incorrect information. However, existing benchmarks primarily emphasize cognitive-level modifications while lacking a focus on deeper meta-cognitive processes. To bridge this gap, we introduce **CogEdit**, a novel benchmark designed to evaluate MLLMs' meta-cognitive knowledge editing abilities across three levels: (1) **Counterfactual-Driven Editing**, assessing self-awareness of knowledge correctness changes; (2) **Boundary Constraint Editing**, ensuring appropriate generalization without unintended interference; and (3) **Noise-Robust Editing**, promoting reflective evaluation of uncertain information. To advance meta-cognitive editing, we propose **MIND** (Meta-cognitive INtegrated Dynamic Knowledge Editing), a framework that constructs a meta-knowledge memory for self-awareness, employs game-theoretic interactions to monitor knowledge activation, and incorporates label refinement for noise-robust updates. Extensive experiments show that **MIND** significantly outperforms existing cognitive editing approaches, achieving strong performance on both traditional and meta-cognitive knowledge editing benchmarks.

\*Equal Contribution.

<sup>†</sup>Project Leader

\*Corresponding Author



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792531>

## CCS Concepts

• Computing methodologies → Computer vision problems.

## Keywords

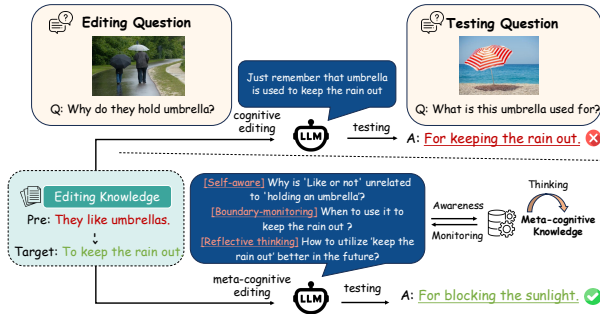
Vision and language, Multimodal understanding, Knowledge Editing

### ACM Reference Format:

Zhaoyu Fan, Kaihang Pan, Mingze Zhou, Bosheng Qin, Juncheng Li, Shengyu Zhang, Wenqiao Zhang, Siliang Tang, Fei Wu, and Yueting Zhuang. 2026. Towards Meta-Cognitive Knowledge Editing for Multimodal LLMs. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792531>

## 1 Introduction

Multimodal Large Language Models (MLLMs) are the engines powering a new generation of web applications, from visual search and content moderation to intelligent virtual assistants. However, because they are typically trained on static web scrapes, these models struggle to keep pace with the dynamic and often chaotic nature of online information. Knowledge Editing [4, 10, 19, 33, 38–41, 53, 54, 59, 60] offers an efficient way for updating large language models (LLMs) to correct errors or outdated information without negative side effects, which has been further extended to the editing of multimodal large language models (MLLMs), known as multimodal editing. Previous research suggests that the process of knowledge editing closely resembles **human cognition** [43, 58]. It enables MLLMs to incorporate new knowledge and rectify their previous misconceptions. To evaluate the effectiveness of this cognitive process, three common metrics are used [7]. The first is **Reliability**, which measures whether the model can accept new



**Figure 1: Difference between Cognitive Editing and Meta-cognitive Editing.** Meta-cognitive editing requires “thinking about thinking”, additionally involving self-awareness, boundary monitoring, reflective thinking.

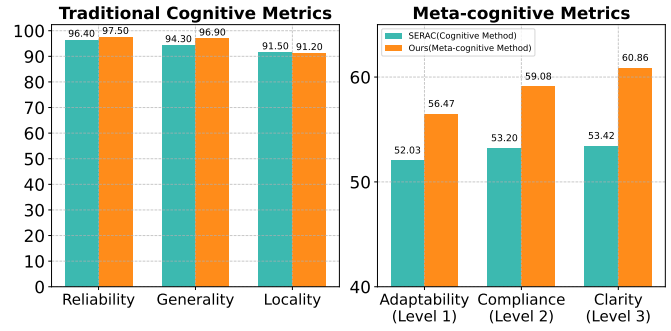
knowledge to achieve the desired editing outcome. The second is **Generality**, which refers to whether the newly acquired knowledge can be transferred to similar scenarios. The third is **Locality**, which requires that unrelated knowledge remains unchanged and is not mistakenly altered during the process.

Ensuring the timeliness and veracity of Web information is critical, yet current cognitive knowledge editing methods remain inadequate. First, these methods perform mechanical replacements without identifying knowledge obsolescence. Second, they process knowledge superficially, failing to comprehend the underlying rationale for correctness. Finally, they lack reflective thinking regarding the editing process itself, which is essential for refining updates amidst noisy and adversarial data.

The difference between cognitive and meta-cognitive editing is shown in Figure 1, to address the limitations of cognitive editing, effective knowledge editing should be **META-COGNITIVE** [13, 23, 50]. Meta-cognition encompasses a model’s ability to assess and modify its own reasoning and learning processes. In contrast to basic cognition, meta-cognition emphasizes the capacity to “think about thinking” and incorporates three additional levels.

- **Level 1: Self-Awareness** [2]. The system, while replacing outdated knowledge with new knowledge, is capable of self-awareness regarding the reasons why the old knowledge was incorrect and needed to be replaced in the corresponding context.
- **Level 2: Boundary Monitoring** [32]. The system is capable of monitoring the utilization boundary of new knowledge, preventing it from being excessively overgeneralized and thereby avoiding the erroneous overshadowing of truly relevant knowledge.
- **Level 3: Reflective Thinking** [45]. The system evaluates its knowledge update processes and refines its adaptation strategies. It assesses external inputs to determine whether they should be accepted and, if so, which portions are relevant for integration.

Existing knowledge editing datasets primarily evaluate a model’s cognitive capabilities but fail to effectively reflect its meta-cognitive abilities. To address this limitation, we introduce **COGEDIT**. It is a novel benchmark designed to assess the meta-cognitive abilities of MLLMs during the knowledge editing process, which encompasses tasks of varying dimensions across diverse domains. In alignment



**Figure 2: The performance of MIND and SERAC [41] on traditional cognitive editing benchmark and our proposed CogEdit.**

with the three key levels outlined earlier, our benchmark evaluates models across three essential criteria:

**Counterfactual-Driven Editing (Level 1)** – When proposing a counterfactual assumption that alters the content of specific knowledge, the model should be **self-aware** that the change in the correctness of the knowledge is attributed to the introduction of the counterfactual condition.

**Boundary-Constraint Editing (Level 2)** – The model should **monitor the utilization boundary** of the new knowledge to prevent it from being overgeneralized to seemingly similar yet unrelated scenarios and mistakenly suppressing knowledge that should be activated.

**Noise-Robust Editing (Level 3)** – When confronted with noisy knowledge of uncertain accuracy, the model must critically assess its validity through **reflective thinking**, determining whether it should replace existing ones.

With our newly designed benchmark, we perform a comprehensive evaluation of SOTA multimodal editing methods for meta-cognitive editing. As shown in Figure 2, in terms of **Adaptability**, **Compliance**, and **Clarity**—the metrics we propose to measure meta-cognitive editing capabilities (detailed in section 3)—though existing methods [41] achieve strong performance on cognitive knowledge editing [8], they lack meta-cognitive capabilities including self-awareness, boundary monitoring, and reflective thinking.

To address these limitations and advance meta-cognitive knowledge editing, we introduce **MIND** (Meta-cognitive INtegrated Dynamic Knowledge Editing), a framework that emulates the human knowledge editing process by mimicking how humans learn new information within their minds.

MIND begins by building a meta-knowledge memory, including both meta-declarative and meta-conditional components, to enable self-aware knowledge updates. It then applies game theory to monitor knowledge activation through old-new knowledge interactions and employs a label refiner to filter noisy inputs, enhancing robustness. This enables MLLMs to perform “thinking about thinking” by regulating their own cognitive processes. As shown in Figure 2, MIND achieves strong performance on both standard and newly proposed meta-cognitive editing benchmarks.

Overall, our main contributions are three-fold:

- We introduce CogEdit, a benchmark to evaluate meta-cognitive knowledge editing capabilities for MLLMs.

- We propose MIND, a framework that simulates human-like meta-cognitive learning with self-awareness, boundary monitoring, and reflective thinking.
- MIND achieves strong performance on both traditional cognitive knowledge editing benchmark and our newly developed meta-cognitive benchmark.

## 2 Related Work

Multimodal Knowledge Editing, a critical task for maintaining the currency and accuracy of MLLMs trained on dynamic web data, is the process of modifying, updating, or correcting specific knowledge within MLLMs [9, 27, 28, 30, 35, 61] while preserving its overall performance across different modalities.

Existing multimodal editing benchmarks [7, 18, 25, 57] evaluate the effectiveness of editing using the Reliability, Generality, and Locality metrics [19]. These metrics assess the cognitive capability of editing methods, which is crucial for web applications like real-time fact-checking on social media or updating product information on e-commerce platforms. However, they exhibit limited diversity. Considering that knowledge editing is inherently meta-cognitive, we propose **CogEdit**, which introduces diversity as an additional metric to evaluate whether an editing method possesses self-awareness, boundary monitoring, and reflective thinking.

From a methodological perspective [43], existing knowledge editing approaches can be categorized into two types: (1) Intrinsic Knowledge Editing [10, 33, 38, 40], which directly modifies the model’s internal parameterized knowledge, and (2) External Knowledge Resorting [4, 41, 53, 59, 60], which preserves the original model parameters while incorporating external knowledge sources through edited samples. However, neither of these approaches exhibits meta-cognitive capabilities, making them less effective in handling the noisy and ever-evolving nature of web-scale information. In contrast, we introduce **MIND**, a knowledge editing framework that simulates a human mind’s meta-cognitive learning process to better address these challenges.

## 3 CogEdit Benchmark

In this section, we propose **CogEdit**, a meta-cognitive multimodal editing benchmark that encompasses **multiple capability levels** across **diverse domains**. Corresponding to the three levels of meta-cognition (i.e., self-awareness, boundary monitoring, reflective thinking), CogEdit evaluates multimodal editing across three essential criteria. We give more details of CogEdit in Appendix A.

*Task Definition.* Assuming we have an MLLM parameterized by  $\theta$  that map the input  $(i_e, x_e)$  to the prediction  $y_o$ , where  $i_e$  refers to the editing image input,  $x_e$  refers to the editing text prompt input and  $y_o$  denote as the origin output. Multimodal editing aims to change prediction from  $y_o$  to  $y_e$  after an updated  $\theta_e$ . On this basis, we introduce our triple-level editing subtasks, as illustrated in Figure 3. A formal definition of the task and corresponding evaluation metrics can be found in Appendix A.

*Level 1: Counterfactual-Driven Editing.* The first level of CogEdit focuses on evaluating the model’s ability to demonstrate **self-awareness**. When a counterfactual assumption alters the correctness of specific knowledge, the model should recognize that the

change in the correctness of that knowledge is due to the introduction of the counterfactual condition. When these conditions are subsequently removed, the prior knowledge should be reactivated.

The task assesses a model’s capability to incorporate counterfactual knowledge by measuring **Fidelity**—the accuracy under a counterfactual assumption—and **Adaptability**—the ability to recover original knowledge once the assumption is removed.

*Level 2: Boundary Constraint Editing.* The second level of CogEdit evaluation assesses the model’s ability of **boundary monitoring**, preventing the new knowledge being overgeneralized to seemingly similar yet unrelated scenarios, and avoiding the erroneous suppression of knowledge that should be activated.

This task evaluates a model’s ability to apply edited knowledge appropriately by measuring **Reliability**—its accuracy in the original context after editing—and **Compliance**—its ability to adapt its responses under boundary constraints without overgeneralizing the edited knowledge.

*Level 3: Noise-Robust Editing.* The third level in CogEdit evaluates whether the model can engage in **reflective thinking**. When confronted with noisy or uncertain knowledge, the model should critically assess its validity and determine whether it should be accepted as a replacement for existing knowledge.

This task measures a model’s ability to extract and retain useful information from noisy updates. Specifically, **Clarity@K** quantifies the model’s robustness by evaluating its accuracy as the level of injected noise increases, with  $K$  denoting the number of distracting elements.

*Data Collection.* To ensure diversity in data sources, we curated a total of 1,174 VQA questions from various multimodal QA datasets [20, 37]. These questions span various specialized domains, including sports, arts, science, everyday activities and so on. The selection process also considered different capability tiers, categorized from simplest to most complex: **Perception**, **Recognition**, **Understanding**, and **Reasoning**. We use open-source models [6, 16, 24, 52] to help us generate and refine the data instances.

## 4 Approach: Meta-Cognitive Editing

We propose a meta-cognitive multimodal editing method, **MIND** (Meta-cognitive **IN**tegrated **DY**namic knowledge editing), to overcome the limitations of cognitive-based editing, as shown in Figure 4. **MIND** retains the activation pathways of the original knowledge during the editing process, guiding the editing and updating based on scene features, and extracting relevant information from the editing examples. It integrates a self-aware meta-cognitive memory(4.1), a game-theoretic monitoring mechanism(4.2), and a meta-label refiner(4.3), enabling meta-cognitive knowledge editing in a manner similar to how the human mind learns and adapts.

### 4.1 Self-Aware Meta Knowledge

*Meta-Memory Construction.* Feed-forward layers in transformer models, which constitute a significant portion of the model’s parameters, function as key-value memory structures [14]. Thus make it possible to edit knowledge through editing feed-forward layers.

Since activation and inhibition mappings influence the model’s activation and forwarding logic, we define a Meta-Memory [3,

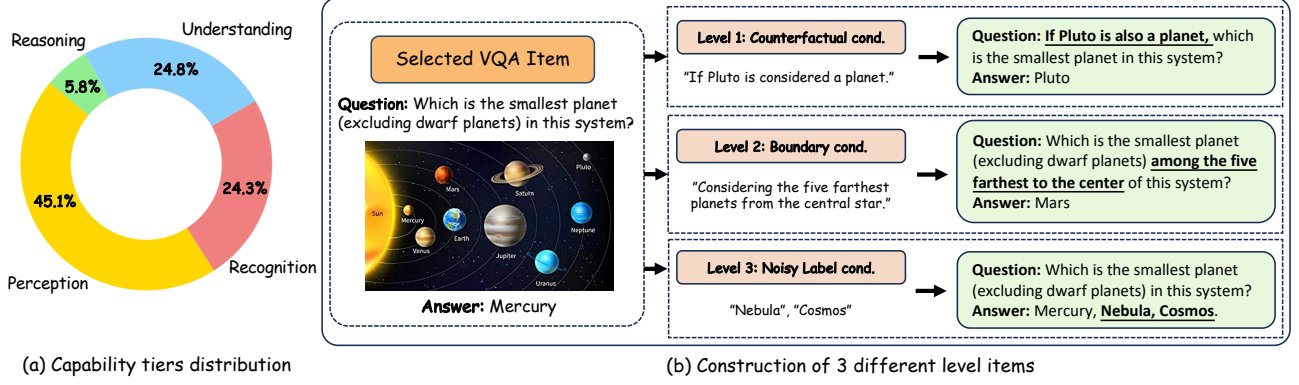


Figure 3: (a) Distribution of problem difficulties in CogEdit. (b) The transformation process from an original VQA instance to edited instances at different levels.

42] projection matrix  $\text{Mem} \in \mathbb{R}^{d' \times d'}$  which consists of the meta-knowledge units  $m_i$ , which is a mapping function that extracts new activation values from original activation values. For a feed-forward layer with an inner hidden dimension of  $d'$ , the meta-memory  $\text{Mem}$  can be represented with:

$$q_{\text{output}} = \text{Mem}(q_{\text{input}}) = \left[ m_i(q_{\text{input}_i}) \right]_{i=0}^{d'-1} \quad (1)$$

$$m_i = \text{split}(\text{Mem})_i = \text{Mem}[:, i] \quad \forall i \in \{0, 1, \dots, d' - 1\} \quad (2)$$

where  $q_i$  and  $q_o$  are original and new activation values of dimension  $d'$ , respectively.

Meta-knowledge is parameterized and stored in meta-memory, and it can be categorized into different levels, including **meta-declarative knowledge**  $m_i^{\text{decl}}$  and **meta-conditional knowledge**  $m_i^{\text{cond}}$ .

**Meta-declarative knowledge** refers to a language model's self-awareness of its knowledge, structure, and limitations. It goes beyond storing facts by allowing the model to assess how reliable its responses are and where they come from. The Meta-declarative knowledge is defined as a one-hot vector  $\mathbf{m}_i^{\text{decl}} = [0, 0, \dots, 1, \dots, 0]^T \in \mathbb{R}^{d'}$ , where the  $i$ -th position is 1, and all other entries are 0.

**Meta-conditional knowledge** refers to a model's awareness of when and under what conditions specific knowledge should be applied. It helps the model adapt by evaluating the relevance and appropriateness of its knowledge in varying contexts. This meta-knowledge helps the model adapt and make informed decisions by guiding when and how to apply information effectively. The Meta-conditional knowledge is defined as a zero vector  $\mathbf{m}_i^{\text{cond}} = \mathbf{0} \in \mathbb{R}^{d'}$ .

Specifically, the **Meta-Memory (Mem)** is intended to be a matrix of size  $[d' \times d']$ , where  $\text{Mem}[:, i]$  refers to the  $i$ -th column vector of the matrix. Each Meta Knowledge Unit corresponds to a column vector of dimension  $[d' \times 1]$ , and is associated with a unique index within the Meta-Memory.

The **Meta-conditional knowledge** and **Meta-declarative knowledge** two vectors together form the Meta Knowledge Unit. A collection of such units composes the entire Meta-Memory. Initially,  $m_i^{\text{decl}} = 1$  if  $i$  equals the target index, and  $m_i^{\text{decl}} = 0$  otherwise; meanwhile,  $m_i^{\text{cond}} = \mathbf{0}$  for all  $i$ , indicating that the meta-memory retains the original declarative knowledge while no conditional knowledge has been learned yet.

The motivation for this initialization design comes in part from psychological theories of meta-knowledge. In particular, meta-declarative knowledge refers to an individual's understanding and evaluation of their own knowledge, suggesting that the initialization should remain neutral and not interfere with the distribution of  $q_r$ . In contrast, meta-conditional knowledge captures how individuals adaptively apply knowledge in various contexts – which is exactly what our editing process aims to model and learn.

From an empirical perspective, we also observed in exploratory experiments that using a diagonal matrix for initialization led to faster convergence and more stable editing. This supports its use as a practical and effective starting point.

## 4.2 Game Theory based Meta-Memory Monitoring

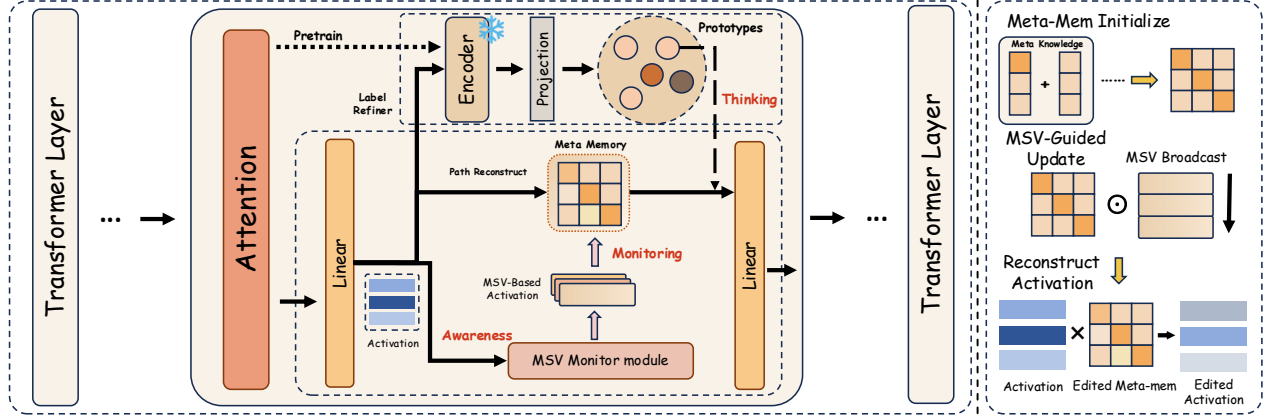
**Shapley Value.** is a fundamental concept in cooperative game theory, used to fairly quantify the marginal contribution of each player within a coalition. Given a set of players  $N = \{1, 2, \dots, n\}$  and a characteristic function  $v : 2^N \rightarrow \mathbb{R}$  that assigns a value to every coalition  $S \subseteq N$ , the Shapley Value of player  $i \in N$  is defined as

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S)). \quad (3)$$

This expression averages the marginal contributions of player  $i$  across all possible coalition formation orders, ensuring a fair allocation of the total value among the players.

Since most knowledge is stored in feed-forward layers, and meta-memory manipulation provides a mechanism to control the knowledge that needs to be activated, it is crucial to understand the importance of each Meta Knowledge Unit  $m_i$  in the hidden layer within a specific knowledge context. To address this, we introduce the concept of the **Meta-memory Shapley Value (MSV)** [47, 55], which quantifies the marginal contribution of each unit to the overall editing performance. In alignment with the intermediate activation values  $q$  between two layers of the FFN, MSV can be defined as follows:

$$\phi(\text{Mem}, q) = \sum_{s \in \text{Mem} \setminus \{i\}} (v(s \cup \{i\}, q) - v(s, q)) \quad (4)$$



**Figure 4: The MIND framework unifies meta-memory, monitoring mechanisms, and label refinement for meta-cognitive multimodal knowledge editing.**

where  $\text{Mem}$  denotes the set of all Meta Knowledge Units, and  $s$  represents a subset of units excluding  $m_i$ . The term  $v(s, q)$  quantifies the performance or contribution of the subset  $s$  of Meta Knowledge Units for the query  $q$ , while  $v(s \cup \{i\}, q)$  denotes the performance or contribution of the subset  $s$  combined with the unit  $m_i$  for the same query.

Inspired by prior works on efficient Shapley value estimation [21, 34, 48], due to the computational complexity of calculating the marginal contribution of each memory unit, we employ an **MSV Monitoring module** with MLP as the approximation of MSV. The MLP is trained to predict the MSV according to the current input query  $q$ . The marginal contribution of the  $i$ -th unit of meta-memory is computed as follows:

$$\phi_i(\text{Mem}, q) \leftarrow \text{Softmax}(\text{Shapley}(m_i, q)) \quad (5)$$

Leveraging the extracted information from MSV facilitates a more effective monitoring of the knowledge update process.

To effectively determine when newly edited knowledge should be applied, MSV is utilized to guide the meta-memory update process, enabling a structured monitoring of meta-memory, which operates as follows:

$$q_r = \left[ (q_i \cdot m_i) \odot \phi_i(\text{Mem}, q) \right]_{i=0}^{d'-1} \quad (6)$$

### 4.3 Reflective based Label Refinement

*Prototype-Based Meta-Label Refiner.* Inspired by [51], we incorporate concepts from the partial label learning framework and introduce a reflective memory module that stores learned prototype knowledge associated with different label categories. This module supports reflective thinking by updating prototype representations based on new labels. Initialized with pre-trained weights, it adaptively refines specific prototypes during editing using edit labels. The encoder processes activation values  $q$  to capture contextual understanding. Using these embeddings, contextual label prototypes are retrieved from the prototype bank  $\mathcal{P} = \{\mathbf{m}_k\}_{k=1}^K$ , where  $\mathbf{m}_k$  represents embeddings encapsulating relevant label-related knowledge. The retrieved contextual label knowledge  $p$  is then projected with a learnable projection matrix  $\mathbf{W}_p$  to get meta-label info, then serves as a shifting vector and is combined with the meta-memory

monitored activation values  $q_r$  to produce the refined representation:

$$q_{\text{refined}} = (1 - \beta)q_r + \beta\mathbf{W}_p p \quad (7)$$

where  $\beta$  controls the contribution of the monitored activation and the meta-label-based shift. The incorporation of reflective thinking ensures that the model does not passively store information but actively refines and adapts its learned representations based on prior knowledge and newly observed samples.

*Supervised Contrastive Pre-Training.* To further improve the robustness of reflective-based label refinement, we employ contrastive learning [22, 49] during the pre-training of the prototype bank  $\mathcal{P}$ . Each training instance consists of a question scenario sampled from either GQA or OK-VQA, its corresponding ground truth label, and a generated negative noisy label that is unrelated to the scenario. During pre-training, the complete Label Refiner module is inserted after the final layer of the backbone model. We use a supervised contrastive loss to optimize  $\mathcal{P}$ : the feature representation of the ground truth label and the question scenario (averaged along the sequence length after being processed by  $\mathcal{P}$ ) are encouraged to be close, while the features of the ground truth label and the noisy label are pushed apart. The resulting representation, referred to as the *meta-label*, encodes label semantics and is designed to be fused with the intermediate activation  $q_r$ . Formally,  $\mathcal{P}$  is parameterized as a  $K \times K$  matrix that captures label-related information, where a larger  $K$  reflects stronger capacity to encode rich label semantics. In the forward pass, the retrieval of label knowledge corresponds to a multiplication with  $\mathcal{P}$ , thereby reinforcing its ability to distinguish between useful and misleading information.

## 5 Experiment

In this section, we first evaluate the editing capability of **MIND** on our meta-cognitive editing benchmark, with results presented in 5.2. We also assess its performance on a traditional cognitive benchmark, as shown in 5.3. These experiments, along with an in-depth analysis 6.1, collectively demonstrate its meta-cognitive ability. Furthermore, we evaluate **MIND** in a lifelong editing scenario in Appendix C.

**Table 1: Main results on CogEdit, including Counterfactual-Driven Editing, Boundary Constraint Editing, and Noise-Robust Editing.**

|                        |                   | COUNTERFACTUAL |               | BOUNDARY EDITING |             | NOISY EDITING |             |
|------------------------|-------------------|----------------|---------------|------------------|-------------|---------------|-------------|
| Method                 |                   | Fidelity↑      | Adaptability↑ | Reliability↑     | Compliance↑ | Clarity@2↑    | Clarity@4↑  |
| MiniGPT-4              |                   |                |               |                  |             |               | Size: 7.3B  |
| Base Methods           | FT (vision block) | 55.08          | 16.72         | 52.56            | 23.20       | 19.29         | 17.84       |
|                        | FT (last layer)   | 67.49          | 15.85         | 60.93            | 24.66       | 17.66         | 14.12       |
| Cognitive Editing      | T-Patcher         | 73.16          | 19.43         | 62.15            | 24.72       | 27.76         | 25.44       |
|                        | IKE               | 99.66          | 13.67         | 99.92            | 15.72       | 14.98         | 10.72       |
|                        | MEND              | 60.55          | 20.88         | 73.92            | 33.10       | 10.43         | 3.50        |
|                        | SERAC             | 99.26          | 29.98         | 99.68            | 40.49       | 30.88         | 30.83       |
|                        | WISE              | 76.29          | 27.63         | 79.33            | 31.00       | 29.33         | 29.01       |
| Meta-Cognitive Editing | MIND (Ours)       | 98.16          | 34.31         | 88.41            | 43.21       | 34.98         | 34.87       |
| LLaVA                  |                   |                |               |                  |             |               | Size: 7.06B |
| Base Methods           | FT (vision block) | 30.97          | 18.60         | 24.27            | 26.47       | 19.50         | 18.95       |
|                        | FT (last layer)   | 71.66          | 28.87         | 45.69            | 21.98       | 21.98         | 18.20       |
| Cognitive Editing      | T-Patcher         | 82.55          | 33.62         | 77.64            | 29.39       | 30.33         | 30.69       |
|                        | IKE               | 78.44          | 31.87         | 76.51            | 28.26       | 34.68         | 39.50       |
|                        | MEND              | 85.36          | 47.44         | 94.90            | 44.84       | 4.98          | 4.98        |
|                        | SERAC             | 98.84          | 52.03         | 99.72            | 53.20       | 53.42         | 47.88       |
|                        | WISE              | 78.72          | 37.39         | 65.67            | 36.23       | 47.66         | 50.28       |
| Meta-Cognitive Editing | MIND (Ours)       | 99.87          | 56.47         | 99.33            | 59.08       | 60.86         | 58.92       |

**Table 2: Main results on cognitive benchmark MMEdit**

| Method                 |                   | EDITING VQA   |                |                |              |              | EDITING IMAGE CAPTION |                |                |              |              |
|------------------------|-------------------|---------------|----------------|----------------|--------------|--------------|-----------------------|----------------|----------------|--------------|--------------|
|                        |                   | Reliability ↑ | T-Generality ↑ | M-Generality ↑ | T-Locality ↑ | M-Locality ↑ | Reliability ↑         | T-Generality ↑ | M-Generality ↑ | T-Locality ↑ | M-Locality ↑ |
| MiniGPT-4              |                   |               |                |                |              |              | Size: 7.3B            |                |                |              |              |
| Base Methods           | FT (vision block) | 36.3          | 0.3            | 10.9           | 100.0        | 9.3          | 3.1                   | 0.0            | 2.2            | 100.0        | 8.6          |
|                        | FT (last layer)   | 70.1          | 65.7           | 63.9           | 72.6         | 65.8         | 67.4                  | 65.1           | 62.8           | 63.5         | 52.7         |
| Cognitive Editing      | T-Patcher         | 83.0          | 68.2           | 66.0           | 84.8         | 82.0         | 83.8                  | 72.3           | 67.7           | 93.9         | 83.6         |
|                        | IKE               | 100.0         | 94.9           | 90.5           | 50.3         | 3.7          | 90.9                  | 81.6           | 88.5           | 52.2         | 4.7          |
|                        | SERAC             | 87.7          | 87.6           | 85.9           | 97.5         | 14.2         | 91.8                  | 91.4           | 91.0           | 97.9         | 7.2          |
|                        | MEND              | 98.8          | 98.6           | 82.2           | 98.2         | 81.1         | 96.6                  | 96.1           | 76.3           | 98.4         | 75.3         |
|                        | WISE              | 91.8          | 90.4           | 87.1           | 99.9         | 86.3         | 95.8                  | 91.7           | 88.8           | 100.0        | 81.5         |
| Meta-Cognitive Editing | MIND(Ours)        | 97.0          | 95.2           | 91.7           | 93.6         | 72.2         | 93.2                  | 90.7           | 81.3           | 99.6         | 84.3         |
| LLaVA                  |                   |               |                |                |              |              | Size: 7.06B           |                |                |              |              |
| Base Methods           | FT (vision block) | 63.4          | 58.6           | 53.7           | 92.5         | 66.7         | 48.3                  | 49.4           | 45.1           | 99.8         | 93.6         |
|                        | FT (last layer)   | 62.5          | 61.9           | 59.9           | 64.8         | 32.9         | 73.4                  | 72.7           | 67.9           | 99.0         | 91.4         |
| Cognitive Editing      | T-Patcher         | 87.8          | 81.5           | 78.3           | 67.5         | 47.6         | 85.3                  | 87.2           | 81.0           | 75.4         | 44.2         |
|                        | IKE               | 95.4          | 86.2           | 79.1           | 52.6         | 17.3         | 92.5                  | 84.3           | 76.6           | 52.5         | 15.6         |
|                        | SERAC             | 96.4          | 94.3           | 87.2           | 91.5         | 23.5         | 97.5                  | 96.9           | 97.6           | 87.6         | 29.2         |
|                        | MEND              | 93.7          | 92.3           | 91.9           | 91.0         | 87.2         | 94.8                  | 92.2           | 89.6           | 91.3         | 90.3         |
|                        | WISE              | 87.0          | 80.3           | 73.5           | 100.0        | 81.5         | 94.6                  | 87.6           | 70.9           | 99.9         | 78.3         |
| Meta-Cognitive Editing | MIND (Ours)       | 97.5          | 96.9           | 92.5           | 91.2         | 76.1         | 96.9                  | 96.8           | 92.7           | 95.6         | 79.2         |

## 5.1 Experiment Setup

**Datasets.** To evaluate the meta-cognitive editing capabilities of the model, we conduct experiments on two benchmarks: CogEdit and MMEdit. These experiments aim to demonstrate that our method exhibits both meta-cognitive and standard cognitive abilities. See more details in Appendix B.

**Baselines.** In this study, we selected several cognitive-based widely adopted multimodal model editing methods as baselines, including Transformer-Patcher [19], IKE [59], MEND [40], SERAC [41], WISE [53], and standard fine-tuning. Experiments were conducted on two distinct MLLM architectures: MiniGPT-4 [61], representing Qformer-based models [29], and LLaVA-v1.5 [35], representing projection-based models.

## 5.2 Results on CogEdit Benchmark

This editing result is shown in Table 1, compared with non-meta-cognitive methods, our meta-cognitive editing method achieves SOTA results on our CogEdit benchmark.

*Limitations of Cognitive-Based Editing Methods.* Existing knowledge editing methods primarily focus on ensuring that the model produces the correct output in edited scenarios. However, these methods fail to enhance the model’s understanding of why a particular output is expected in a given context, **indicating a lack of meta-cognitive ability**. As shown in Table 1, we observe the following: **(1) In counterfactual scenarios**, T-Patcher, IKE, and MEND exhibit generally low Adaptability, suggesting that although these methods may acquire knowledge in counterfactual settings, the newly edited knowledge interferes with the application of prior knowledge once the counterfactual condition is removed. This indicates a lack of self-awareness in recognizing when outdated knowledge should be discarded. **(2) In boundary scenarios**, while cognitive-based methods may achieve relatively high Reliability, they generally perform poorly in terms of Compliance. This suggests that these methods fail to determine when the newly edited knowledge should be applied, highlighting a deficiency in boundary monitoring. **(3) In noisy scenarios**, although T-Patcher, IKE, and WISE are able to perform edits on examples containing noise. However, their lower **Clarity** scores suggest that these methods struggle to isolate informative signals from noisy labels, reflecting a deficiency in reflective reasoning capabilities.

*Advantages of MIND.* Since the cognitive-based editing module struggles to effectively extract useful information from editing samples, MIND demonstrates significantly improved performance under such conditions. **(1) Enhanced self-awareness:** In counterfactual scenarios, MIND achieves higher Adaptability than cognitive-based methods, indicating its ability to better recognize when to discard outdated knowledge, demonstrating its superior capability of self-awareness. **(2) Stronger boundary monitoring:** In boundary-constrained scenarios, MIND attains higher Compliance than cognitive-based methods, showcasing its capability to accurately determine when to apply newly acquired knowledge, demonstrating its boundary monitoring ability. **(3) Improved reflective thinking:** In noisy editing scenarios, MIND achieves higher Clarity than cognitive-based methods, highlighting its ability to analyze and extract useful information from noisy labels, demonstrating its reflective thinking ability.

## 5.3 Results on the MMEdit Dataset

In addition to evaluating meta-cognitive editing, we assess our method on the widely used cognitive-based editing dataset, MMEdit [7], as shown in Table 2. Compared to the reported multimodal knowledge editing results, our method also achieves competitive performance. We observe that: **(1)** Most of the cognitive-based methods can not effectively balance Reliability, Generality, and Locality. **(2)** While cognitive-based methods such as MEND and WISE achieve high Reliability, they often do so at the expense of Generality, leading to a reduction in the generalizability of the edited knowledge. **(3)** For the Locality metric, methods such as In-Context Editing and SERAC tend to perform suboptimally. In contrast, MIND controls

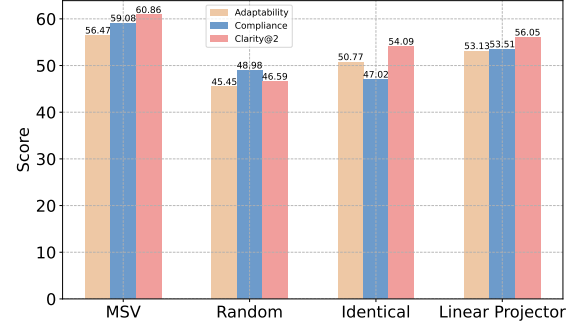


Figure 5: Effectiveness of MSV-Monitored Update

the activation of the meta-knowledge unit to regulate when both prior and newly acquired knowledge should be activated, thereby achieving a balance across multiple metrics. This ensures effective meta-cognitive editing while preserving cognitive ability.

## 6 Results on Lifelong Editing Scenario

Lifelong model editing [17, 53] aims to edit the specific model over and over again, which enables large language models to continuously update and adapt their knowledge over time without requiring full retraining.

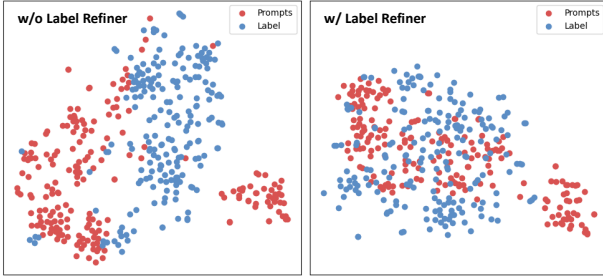
We also evaluated the effectiveness of our method in a lifelong editing scenario. To ensure fairness and eliminate task-induced biases, experiments were conducted on MMEdit (excluding IKE, as lifelong editing is not meaningful for IKE), with results presented in Table 5. Compared to cognitive-based methods, MIND also demonstrates strong performance. We observe that: **(1)** Most methods, such as T-Patcher and MEND, exhibit a significant decline in all metrics after multiple editing iterations. However, as  $T$  increases, the rate of decline slows. In contrast, MIND maintains relatively high editing performance even as  $T$  increases. **(2)** Editing methods specifically designed for lifelong learning, such as WISE, achieve high Locality scores. However, they fail to maintain consistently high Reliability and Generality. MIND is able to sustain relatively superior performance compared to other methods, indicating that it can continuously learn meta-knowledge from ongoing edits while maintaining effectiveness.

### 6.1 In Depth Analysis

*Effect of Individual Components.* We evaluate the effectiveness of our approach across different component combinations. **(1) Only MSV monitor module:** Freezing meta-memory parameters forced reliance solely on MSV constraints. Consequently, the meta-knowledge lacked sufficient knowledge-related features, leading to suboptimal performance. **(2) Using only the meta-memory:** While the model could redistribute FFN activation values to learn editing-relevant knowledge, it failed to discern when to utilize new versus outdated information, hampering adaptability and compliance. **(3) Meta-memory and MSV monitor module:** The model effectively updated meta-memory parameter, however, without the label refiner, extracting useful features remained difficult, resulting in low

**Table 3: Ablation study of our method on CogEdit**

| Method                         | COUNTERFACTUAL |               | BOUNDARY EDITING |              | NOISY EDITING |              |
|--------------------------------|----------------|---------------|------------------|--------------|---------------|--------------|
|                                | Fidelity↑      | Adaptability↑ | Reliability↑     | Compliance↑  | Clarity@2↑    | Clarity@4↑   |
| <i>only</i> MSV monitor module | 76.39          | 39.21         | 79.52            | 44.73        | 31.72         | 26.90        |
| <i>only</i> inner meta-mem     | 97.91          | 47.31         | 97.89            | 42.69        | 54.46         | 52.27        |
| Meta-mem + MSV module          | 91.15          | 50.28         | 93.44            | 49.77        | 56.27         | 50.88        |
| <i>only</i> label refiner      | 80.99          | 48.69         | 93.66            | 58.71        | 36.12         | 33.63        |
| MIND(Ours)                     | <b>99.87</b>   | <b>56.47</b>  | <b>99.33</b>     | <b>59.08</b> | <b>60.86</b>  | <b>57.40</b> |

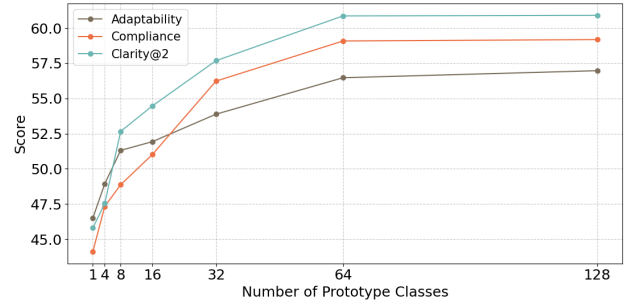
**Figure 6: Ablation study on meta-label refiner: Alignment Visualization**

knowledge clarity. (4) **Only label refiner**: Merely enhancing label features without the capability to redistribute FFN activations proved insufficient to control neural pathways, causing a significant decline in success rates under counterfactual and boundary scenarios.

*Effect of MSV monitor module.* We replace our MSV monitor module by (1) **Random distribution**; (2) **Identical distribution**: means each meta-knowledge unit owns the same significance; (3) **Linear Projector**: as the simplified version of our module. Various modules are employed to monitor meta-memory updates and assess the effectiveness. Figure 5 demonstrates that our MSV monitor module outperforms other approaches, indicating its effectiveness in monitoring meta-memory and guiding the learning process.

*Effect of the Meta-Label Refiner.* To demonstrate the refiner’s information extraction capability, we visualized the hidden state distributions using t-SNE [36]. As shown in Figure 6, without the meta-label refiner, the model struggles to extract features from noisy labels, resulting in label-prompt misalignment. Conversely, the encoder improves alignment and prediction accuracy.

We also study the impact of the number of prototype classes, as shown in Figure 7. Increasing the number of prototypes enables more precise contextual features but incurs higher computational cost. Conversely, too few prototypes result in less accurate contextual encoding and weaker editing effects. When the number of prototypes is sufficiently large (e.g.,  $K = 128$ ), further increases bring marginal performance gains while increasing computational overhead. Therefore, we choose  $K = 64$  to balance accuracy and efficiency.

**Figure 7: Ablation study on meta-label refiner: Effect of Prototype Classes**

## 7 Conclusion

In this paper, we identify the limitations of existing cognitive-based knowledge editing datasets and propose **CogEdit**, a benchmark designed to assess the meta-cognitive capabilities of multimodal knowledge editing. We find that though current knowledge editing methods perform well in traditional editing scenarios, they fail to exhibit meta-cognitive abilities. On this basis, we introduce **MIND**, a novel meta-cognitive knowledge editing approach. Extensive Experimental results demonstrate the effectiveness of MIND, which achieves superior performance on both cognitive and meta-cognitive multimodal editing benchmarks.

## 8 Limitations

Despite the promising results demonstrated in our study, several limitations remain that warrant further investigation.

First, we focus solely on assessing the model’s comprehension of edited multimodal knowledge, without testing its performance in more complex scenarios like image generation and video understanding [11, 26, 31, 44, 46, 56]. Future work should explore how these edits affect results in such cases.

Second, due to computational constraints, we were unable to conduct experiments on large-scale models, such as the 65B LLaMA Adapter V2 [12]. Scaling our method to such models poses additional challenges and is left for future exploration.

## Acknowledgments

This work was supported by the NSFC (62272411), the Key R&D Projects in Zhejiang Province (No. 2025C01030), the Zhejiang NSF (LRG25F020001).

## References

- [1] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. 2015. Vqa: Visual question answering. In *Proceedings of the IEEE international conference on computer vision*. 2425–2433.
- [2] Sarah-Jayne Blakemore and Chris Frith. 2003. Self-awareness and action. *Current opinion in neurobiology* 13, 2 (2003), 219–224.
- [3] John C Cavanaugh and Marion Perlmutter. 1982. Metamemory: A critical examination. *Child development* (1982), 11–28.
- [4] Qizhou Chen, Chengyu Wang, Dakan Wang, Taolin Zhang, Wangyue Li, and Xiaofeng He. 2025. Lifelong knowledge editing for vision language models with low-rank mixture-of-experts. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 9455–9466.
- [5] Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Dollár, and C Lawrence Zitnick. 2015. Microsoft coco captions: Data collection and evaluation server. *arXiv preprint arXiv:1504.00325* (2015).
- [6] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, Bin Li, Ping Luo, Tong Lu, Yu Qiao, and Jifeng Dai. 2024. InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks. *arXiv:2312.14238* [cs.CV] <https://arxiv.org/abs/2312.14238>
- [7] Siyuan Cheng, Bozhong Tian, Qingbin Liu, Xi Chen, Yongheng Wang, Huajun Chen, and Ningyu Zhang. 2023. Can We Edit Multimodal Large Language Models?. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6–10, 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, 13877–13888. <https://aclanthology.org/2023.emnlp-main.856>
- [8] Siyuan Cheng, Bozhong Tian, Qingbin Liu, Xi Chen, Yongheng Wang, Huajun Chen, and Ningyu Zhang. 2023. Can we edit multimodal large language models? *arXiv preprint arXiv:2310.08475* (2023).
- [9] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. 2023. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. *arXiv:2305.06500* [cs.CV] <https://arxiv.org/abs/2305.06500>
- [10] Nicola De Cao, Wilker Aziz, and Ivan Titov. 2021. Editing factual knowledge in language models. *arXiv preprint arXiv:2104.08164* (2021).
- [11] Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. 2025. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 24108–24118.
- [12] Peng Gao, Jiaming Han, Renrui Zhang, Ziyi Lin, Shijie Geng, Aojun Zhou, Wei Zhang, Pan Lu, Conghui He, Xiangyu Yue, Hongsheng Li, and Yu Qiao. 2023. LLaMA-Adapter V2: Parameter-Efficient Visual Instruction Model. *arXiv preprint arXiv:2304.15010* (2023).
- [13] Petros Georgiades. 2004. From the general to the situated: Three decades of metacognition. *International journal of science education* 26, 3 (2004), 365–383.
- [14] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. 2020. Transformer feed-forward layers are key-value memories. *arXiv preprint arXiv:2012.14913* (2020).
- [15] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6904–6913.
- [16] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).
- [17] Tom Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. 2023. Aging with grace: Lifelong model editing with discrete key-value adaptors. *Advances in Neural Information Processing Systems* 36 (2023), 47934–47959.
- [18] Han Huang, Haitian Zhong, Tao Yu, Qiang Liu, Shu Wu, Liang Wang, and Tieniu Tan. 2024. VLKEB: A Large Vision-Language Model Knowledge Editing Benchmark. *arXiv:2403.07350*
- [19] Zeyu Huang, Yikang Shen, Xiaofeng Zhang, Jie Zhou, Wenge Rong, and Zhang Xiong. 2023. Transformer-patcher: One mistake worth one neuron. *arXiv preprint arXiv:2301.09785* (2023).
- [20] Drew A Hudson and Christopher D Manning. 2019. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 6700–6709.
- [21] Cosimo Izzo, Aldo Lipani, Ramin Okhrati, and Francesca Medda. 2020. A baseline for shapley values in MLPs: From missingness to neutrality. *arXiv preprint arXiv:2006.04896* (2020).
- [22] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. 2020. Supervised Contrastive Learning. In *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (Eds.), Vol. 33. Curran Associates, Inc., 18661–18673. [https://proceedings.neurips.cc/paper\\_files/paper/2020/file/d89a66c7c80a29b1bdbab0f2a1a94af8-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2020/file/d89a66c7c80a29b1bdbab0f2a1a94af8-Paper.pdf)
- [23] Emily R Lai. 2011. Metacognition: A literature review. (2011).
- [24] Feng Li, Renrui Zhang, Hao Zhang, Yuanhan Zhang, Bo Li, Wei Li, Zejun Ma, and Chunyuan Li. 2024. Llava-next-interleave: Tackling multi-image, video, and 3d in large multimodal models. *arXiv preprint arXiv:2407.07895* (2024).
- [25] Jiaqi Li, Miaoqiang Du, Chuanyi Zhang, Yongrui Chen, Nan Hu, Guilin Qi, Haiyun Jiang, Siyuan Cheng, and Bozhong Tian. 2024. Mike: A new benchmark for fine-grained multimodal entity knowledge editing. *arXiv preprint arXiv:2402.14835* (2024).
- [26] Juncheng Li, Minghe Gao, Xiangnan He, Siliang Tang, weishi Zheng, Jun Xiao, Meng Wang, Tat-Seng Chua, and Yueting Zhuang. 2026. Mometor++: Advancing Video Large Language Models with Fine-Grained Long Video Reasoning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026).
- [27] Juncheng Li, Minghe Gao, Siliang Tang, Longhui Wei, Jun Xiao, Fei Wu, Richang Hong, Meng Wang, and Qi Tian. 2025. Structure-Induced Gradient Regulation for Generalizable Vision-Language Models. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [28] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. *arXiv:2301.12597* [cs.CV] <https://arxiv.org/abs/2301.12597>
- [29] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*. PMLR, 19730–19742.
- [30] Juncheng Li, Kaihang Pan, Zhiqi Ge, Minghe Gao, Wei Ji, Wenqiao Zhang, Tat-Seng Chua, Siliang Tang, Hanwang Zhang, and Yueting Zhuang. 2023. Fine-tuning multimodal llms to follow zero-shot demonstrative instructions. In *The Twelfth International Conference on Learning Representations*.
- [31] Juncheng Li, Siliang Tang, Linchao Zhu, Wenqiao Zhang, Yi Yang, Tat-Seng Chua, and Fei Wu. 2023. Variational Cross-Graph Reasoning and Adaptive Structured Semantics Learning for Compositional Temporal Grounding. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).
- [32] Moxin Li, Yong Zhao, Yang Deng, Wenxuan Zhang, Shuaiyi Li, Wenya Xie, See-Kiong Ng, and Tat-Seng Chua. 2024. Knowledge Boundary of Large Language Models: A Survey. *arXiv preprint arXiv:2412.12472* (2024).
- [33] Xiaopeng Li, Shasha Li, Shezheng Song, Jing Yang, Jun Ma, and Jie Yu. 2024. Pmet: Precise model editing in a transformer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 18564–18572.
- [34] Tianwei Lin, Jiang Liu, Wenqiao Zhang, Zhaocheng Li, Yang Dai, Haoyuan Li, Zhelun Yu, Wangui He, Juncheng Li, Hao Jiang, et al. 2024. Teamlor: Boosting low-rank adaptation with expert collaboration and competition. *arXiv preprint arXiv:2408.09856* (2024).
- [35] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual Instruction Tuning.
- [36] Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, Nov (2008), 2579–2605.
- [37] Kenneth Marino, Mohammad Rastegari, Ali Farhadi, and Roozbeh Mottaghi. 2019. OK-VQA: A Visual Question Answering Benchmark Requiring External Knowledge. *arXiv:1906.00067* [cs.CV] <https://arxiv.org/abs/1906.00067>
- [38] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in gpt. *Advances in neural information processing systems* 35 (2022), 17359–17372.
- [39] Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2022. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229* (2022).
- [40] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D Manning. 2021. Fast model editing at scale. *arXiv preprint arXiv:2110.11309* (2021).
- [41] Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D Manning, and Chelsea Finn. 2022. Memory-based model editing at scale. In *International Conference on Machine Learning*. PMLR, 15817–15831.
- [42] Thomas O Nelson. 1990. Metamemory: A theoretical framework and new findings. In *Psychology of learning and motivation*. Vol. 26. Elsevier, 125–173.
- [43] Kaihang Pan, Zhaoyu Fan, Juncheng Li, Qifan Yu, Hao Fei, Siliang Tang, Richang Hong, Hanwang Zhang, and Qianru Sun. 2024. Towards unified multimodal editing with enhanced knowledge collaboration. *Advances in Neural Information Processing Systems* 37 (2024), 110290–110314.
- [44] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* 1, 2 (2022), 3.
- [45] Carol Rodgers. 2002. Defining reflection: Another look at John Dewey and reflective thinking. *Teachers college record* 104, 4 (2002), 842–866.
- [46] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- [47] Alvin E Roth. 1988. Introduction to the Shapley value. *The Shapley value* 1 (1988).

- [48] Benedek Rozemberczki, Lauren Watson, Péter Bayer, Hao-Tsung Yang, Olivér Kiss, Sebastian Nilsson, and Rik Sarkar. 2022. The shapley value in machine learning. In *The 31st International Joint Conference on Artificial Intelligence and the 25th European Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, 5572–5579.
- [49] Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola. 2020. What makes for good views for contrastive learning? *Advances in neural information processing systems* 33 (2020), 6827–6839.
- [50] Marcel VJ Veenman, Bernadette HAM Van Hout-Wolters, and Peter Afflerbach. 2006. Metacognition and learning: Conceptual and methodological considerations. *Metacognition and learning* 1 (2006), 3–14.
- [51] Haobo Wang, Ruixuan Xiao, Yixuan Li, Lei Feng, Gang Niu, Gang Chen, and Junbo Zhao. 2022. Pico+: Contrastive label disambiguation for robust partial label learning. *arXiv preprint arXiv:2201.08984* (2022).
- [52] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. 2024. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191* (2024).
- [53] Peng Wang, Zexi Li, Ningyu Zhang, Ziwen Xu, Yunzhi Yao, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. 2024. WISE: Rethinking the Knowledge Memory for Lifelong Model Editing of Large Language Models. arXiv:2405.14768 [cs.CL] <https://arxiv.org/abs/2405.14768>
- [54] Renzhi Wang and Piji Li. 2024. Lemoe: Advanced mixture of experts adaptor for lifelong model editing of large language models. *arXiv preprint arXiv:2406.20030* (2024).
- [55] Eyal Winter. 2002. The shapley value. *Handbook of game theory with economic applications* 3 (2002), 2025–2054.
- [56] Hang Zhang, Xin Li, and Lidong Bing. 2023. Video-LLaMA: An Instruction-tuned Audio-Visual Language Model for Video Understanding. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. 543–553.
- [57] Junzhe Zhang, Huixuan Zhang, Xunjian Yin, Baizhou Huang, Xu Zhang, Xinyu Hu, and Xiaojun Wan. 2024. MC-MKE: A Fine-Grained Multimodal Knowledge Editing Benchmark Emphasizing Modality Consistency. arXiv:2406.13219 [cs.CV] <https://arxiv.org/abs/2406.13219>
- [58] Ningyu Zhang, Yunzhi Yao, Bozhong Tian, Peng Wang, Shumin Deng, Mengru Wang, Zekun Xi, Shengyu Mao, Jintian Zhang, Yuansheng Ni, et al. 2024. A comprehensive study of knowledge editing for large language models. *arXiv preprint arXiv:2401.01286* (2024).
- [59] Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. 2023. Can we edit factual knowledge by in-context learning? *arXiv preprint arXiv:2305.12740* (2023).
- [60] Zexuan Zhong, Zhengxuan Wu, Christopher D Manning, Christopher Potts, and Danqi Chen. 2023. Mquake: Assessing knowledge editing in language models via multi-hop questions. *arXiv preprint arXiv:2305.14795* (2023).
- [61] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. 2023. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592* (2023).

## A Dataset Details

### A.1 Capability tiers in CogEdit Benchmark

Data in CogEdit are considered to own different capability tiers, from low to high, the number of instances are shown in Table 4

- **Perception:** Scene understanding, instance identity, attribute recognition, and spatial localization.
- **Multimodal Recognition:** Text recognition, object recognition.
- **Multimodal Understanding:** Visual reasoning and visual referring expressions.
- **Cascading Reasoning:** Logical reasoning, numerical reasoning, and visual mathematics.

To ensure the reliability and accuracy of the curated data, we conducted a two-pass quality control process. For the first pass, we utilized open-source models [6, 16, 24, 52] to automatically validate and refine the data. In the second pass, a manual review was performed to further ensure correctness in order to remove ambiguous

or poorly phrased entries. This combination of automated and manual methods ensured consistency with the intended domain and capability tiers.

| Capability | Perception | Recognition | Understanding | Reasoning |
|------------|------------|-------------|---------------|-----------|
| Number     | 530        | 285         | 291           | 68        |

**Table 4: Distribution of Instances Across Capability Tiers**

### A.2 Formal Task Definitions and Metrics in CogEdit

**Level 1: Counterfactual condition-driven editing.** For counterfactual condition-driven editing, we evaluate the editing result with two metrics: **Fidelity**, which measures the success rate of editing in counterfactual scenarios, and **Adaptability**, which evaluates the model’s success rate in answering factual questions after the counterfactual edit.

Formally, for a given instance  $(i_e, x_e)$  (e.g.,  $i_e$  is an image of “Solar System”;  $x_e$  = “Which is the smallest planet in this system?”) with original correct answer  $y = \text{“Mercury”}$ , we introduce a counterfactual assumption  $cf_e = \text{“Assume that Pluto is a planet, rather than a dwarf planet”}$ , which adjusts the correct answer to  $y_e = \text{“Pluto”}$  with the editing objective as  $1_{f(cf_e, i_e, x_e; \theta_e(cf_e, i_e, x_e)) = y_e}$ <sup>1</sup> (we use **Fidelity** as the editing success rate). After editing, we remove the counterfactual assumption with  $y_o$  restored to the correct answer, assessing whether the edited model can recall its prior correct knowledge when the assumption no longer holds:  $1_{f(i_e, x_e; \theta_e(cf_e, i_e, x_e)) = y}$  (where **Adaptability** quantifies the success rate of answering correctly after removing the counterfactual assumption).

**Fidelity.** Editing fidelity refers to the ability to change the prediction to  $y_e$ . Therefore, our goal is to update the model’s original parameters  $\theta_o$  to  $\theta_e$ , such that  $f(cf_e, i_e, x_e; \theta_e) = y_e$ . The success rate of editing in a counterfactual scenario is described as:

$$\mathcal{M}_{fidelity} = \mathbb{E}_{(cf_e, i_e, x_e) \sim C(i_e, x_e)} \left[ 1_{f(cf_e, i_e, x_e; \theta_e(cf_e, i_e, x_e)) = y_e} \right] \quad (8)$$

where  $C(i_e, x_e)$  denotes the distribution of counterfactual instances derived from  $(i_e, x_e)$ ,  $y_e$  is the correct answer of the question  $(cf_e, i_e, x_e)$  with counterfactual assumptions.

**Adaptability.** Editing adaptability is used to test whether the model has retained its prior correct knowledge and can revert to the original prediction when the counterfactual assumption is removed. Adaptability is described by the following:

$$\mathcal{M}_{adaptability} = \mathbb{E}_{(i_e, x_e) \sim \mathcal{D}_{base}} \left[ 1_{f(i_e, x_e; \theta_e(cf_e, i_e, x_e)) = y} \right] \quad (9)$$

where  $\theta_e$  refers to the edited parameters,  $y$  is the correct answer of the question  $(i_e, x_e)$ .

<sup>1</sup>1. denotes the indicator function: it returns 1 if the condition is true, 0 otherwise.

**Level 2: Boundary-restricted Editing.** For boundary-restricted editing, we evaluate the editing performance using two metrics: **Reliability**, which measures the success rate of editing under the original, unaltered context, and **Compliance**, which assesses the model’s ability to appropriately constrain the activation of edited knowledge when additional boundary conditions are introduced. Reliability reflects whether the model has correctly internalized the intended factual update, while Compliance tests whether the model can generalize this knowledge appropriately without over-generalization or misuse in contextually incompatible settings.

Formally, for the previously given instance  $(i_e, x_e)$  with editing target  $y_e = \text{“Mercury”}$ , we conduct the conventional editing process to let the model remember the correct answer in the original context, with the editing objective as  $1_{f(i_e, x_e; \theta_e(i_e, x_e)) = y_e}$  (we use **Reliability** as the editing success rate). After editing, we construct a new problem instance by introducing boundary constraints  $b_e = \text{“Considering the five farthest planets from the central star”}$ , which limits the available candidate planets and adjusts the correct answer to  $y'_e = \text{“Mars”}$ , in order to assess whether the edited model knows the boundary to utilize the edited knowledge rather than indiscriminately activating it:  $1_{f(b_e, i_e, x_e; \theta_e(i_e, x_e)) = y'_e}$  (where **Compliance** quantifies the success rate of answering correctly after adding the boundary constraint assumption).

**Reliability.** Reliability in this context refers to the model’s ability to edit successfully based on the original knowledge when applied in the original scenario. It quantifies the model’s success rate in providing accurate responses using the edited parameters  $\theta_e$ . The reliability metric can be defined as follows:

$$\mathcal{M}_{\text{reliability}} = \mathbb{E}_{(i_e, x_e) \sim \mathcal{D}_{\text{base}}} [1_{f(i_e, x_e; \theta_e(i_e, x_e)) = y_e}] \quad (10)$$

Where  $y_e$  is the correct answer of the question  $(i_e, x_e)$ .

**Compliance.** Compliance measures the model’s success in correctly identifying when the new knowledge should be applied, ensuring that the modified response aligns with the constraints of the new scenario. This metric evaluates the model’s ability to know when the new knowledge is applicable. The Compliance metric is defined as:

$$\mathcal{M}_{\text{compliance}} = \mathbb{E}_{(b_e, i_e, x_e) \sim \mathcal{C}(i_e, x_e)} [1_{f(b_e, i_e, x_e; \theta_e(i_e, x_e)) = y'_e}] \quad (11)$$

Where  $y'_e$  is the correct answer of the boundary restricted question  $(b_e, i_e, x_e)$ .

**Level 3: Noisy Label Editing.** To systematically evaluate the model’s adaptability to noisy inputs, we propose a new evaluation metric, **Clarity@K**, which quantifies the model’s ability to maintain accuracy as noise increases.

Formally, for a the same instance  $(i_e, x_e)$  with originally correct answer  $y = \text{“Mercury”}$ , we introduce a perturbation  $\text{noise}_e = \text{“Nebula, Cosmos.”}$ , which are some non-planet labels, to the originally correct prediction, which adjusts the correct answer to  $y_e$  containing noise. After editing, the noisy perturbation is removed to assess whether the edited model can get to know the useful part in edited knowledge:  $1_{f(i_e, x_e; \theta_e(\text{noise}_e, i_e, x_e)) = y}$  (where **Clarity@K** quantifies the model’s ability to maintain accuracy as noise increases and  $K$  denotes the number of noisy elements).

**Clarity@K.** Editing clarity is defined to quantify the editing success rate after editing with the presence of noisy labels. Higher values of **Clarity@K** indicate the model can clearly discern and edit the target knowledge despite the noise. The **Clarity@K** metric is defined as:

$$\mathcal{M}_{\text{Clarity@K}} = \mathbb{E}_{(i_e, x_e) \sim \mathcal{D}_{\text{base}}} [1_{f(i_e, x_e; \theta_e(\text{noise}_e, i_e, x_e)) = y}] \quad (12)$$

Where  $y$  is the correct answer of the question  $(i_e, x_e)$ .

## B Experimental Details

### B.1 Cognitive Dataset Details

**MMEdit.** MMEdit [7] has been selected as the representative of the cognitive-based multimodal knowledge editing benchmark, which comprises two subtasks: Editing VQA (E-VQA) [1] and Editing Image Captioning (E-IC), with data selected from VQAv2 [15] and COCO Caption [5], respectively. These tasks aim to assess a model’s ability to update its understanding of visual and textual inputs while ensuring stability and generalization across different modalities. MMEdit introduces a suite of metrics to evaluate the reliability, locality, and generality of model edits in multimodal settings. Each edit sample includes five examples that measure the following metrics. Reliability measures the accuracy of editing target inputs, locality evaluates the model’s stability on unrelated knowledge (both textual and multimodal), and generality assesses the model’s ability to generalize to rephrased inputs (textual and visual).

### B.2 Baseline Methods

**Fine-tune.** Fine-tuning is one of the simplest methods for adapting models to new knowledge. In our experiments, we conducted fine-tuning on the **last layer** of the MLLM and the **vision block** (vision projector for LLaVA and Qformer for MiniGPT4 respectively) separately.

**Transformer-Patcher.** [19] is an editing method that enables continuous correction of mistakes in Transformer-based language models. The method adds and trains a few neurons (patches) in the last Feed-Forward Network (FFN) layer to correct specific errors without affecting overall model performance.

**MEND.** (Model Editor Networks with Gradient Decomposition) [40] is a scalable method for efficiently editing large pre-trained neural network models using a single input-output pair. By leveraging the low-rank structure of fine-tuning gradients, MEND trains lightweight auxiliary networks to transform these gradients into targeted parameter updates. This approach enables fast, local, and reliable edits to model behavior without significantly affecting unrelated inputs.

**SERAC.** (Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model) [41] is a knowledge editing method requiring external memory. SERAC stores edits in an external memory and uses a scope classifier and counterfactual model to determine when and how to apply these edits.

**IKE.** (In-context Knowledge Editing) [59] explores the potential of in-context learning (ICL) for editing factual knowledge in large

**Table 5: Lifelong editing results of LLaVA on MMEdit for different values of  $T$  ( $T$ : Number of Edits.)**

| Method                 |                   | $T = 10$               |                         |                         |                       |                       | $T = 100$              |                         |                         |                       |                       |
|------------------------|-------------------|------------------------|-------------------------|-------------------------|-----------------------|-----------------------|------------------------|-------------------------|-------------------------|-----------------------|-----------------------|
|                        |                   | Reliability $\uparrow$ | T-Generality $\uparrow$ | M-Generality $\uparrow$ | T-Locality $\uparrow$ | M-Locality $\uparrow$ | Reliability $\uparrow$ | T-Generality $\uparrow$ | M-Generality $\uparrow$ | T-Locality $\uparrow$ | M-Locality $\uparrow$ |
| EDITING VQA            |                   |                        |                         |                         |                       |                       |                        |                         |                         |                       |                       |
| Base Methods           | FT (vision block) | 49.26                  | 46.01                   | 42.13                   | 100.00                | 30.42                 | 41.65                  | 42.53                   | 30.41                   | 98.92                 | 26.48                 |
|                        | FT (last layer)   | 53.84                  | 50.29                   | 45.03                   | 91.61                 | 63.45                 | 39.09                  | 34.06                   | 38.07                   | 92.84                 | 57.04                 |
| Cognitive Editing      | T-Patcher         | 82.16                  | 76.45                   | 76.55                   | 71.10                 | 74.50                 | 70.57                  | 63.39                   | 61.12                   | 36.42                 | 48.90                 |
|                        | SERAC             | 88.09                  | 83.40                   | 83.57                   | 64.91                 | 15.50                 | 88.08                  | 81.53                   | 82.48                   | 62.13                 | 12.90                 |
|                        | MEND              | 22.58                  | 22.30                   | 21.48                   | 32.40                 | 18.22                 | 11.43                  | 19.66                   | 19.31                   | 8.95                  | 9.37                  |
|                        | WISE              | 55.04                  | 52.71                   | 70.99                   | 100.0                 | 92.17                 | 27.53                  | 27.54                   | 21.63                   | 100.0                 | 83.48                 |
| Meta-Cognitive Editing | <b>MIND(Ours)</b> | 84.88                  | 84.44                   | 78.82                   | 91.00                 | 75.23                 | 78.49                  | 70.79                   | 65.60                   | 88.72                 | 73.34                 |
| EDITING IMAGE CAPTION  |                   |                        |                         |                         |                       |                       |                        |                         |                         |                       |                       |
| Base Methods           | FT (vision block) | 46.12                  | 47.56                   | 43.20                   | 100.00                | 84.94                 | 45.96                  | 47.50                   | 42.93                   | 99.9                  | 71.05                 |
|                        | FT (last layer)   | 69.52                  | 68.24                   | 63.47                   | 93.69                 | 74.71                 | 65.14                  | 60.11                   | 56.23                   | 81.05                 | 70.09                 |
| Cognitive Editing      | T-Patcher         | 79.94                  | 80.42                   | 74.52                   | 54.92                 | 68.06                 | 59.88                  | 62.81                   | 54.37                   | 26.08                 | 39.99                 |
|                        | SERAC             | 56.57                  | 56.88                   | 52.62                   | 59.96                 | 14.66                 | 53.35                  | 53.70                   | 49.41                   | 48.04                 | 17.25                 |
|                        | MEND              | 66.49                  | 64.77                   | 58.72                   | 87.25                 | 86.15                 | 56.83                  | 56.89                   | 53.07                   | 87.63                 | 84.64                 |
|                        | WISE              | 73.10                  | 71.21                   | 64.79                   | 88.46                 | 91.80                 | 60.56                  | 57.10                   | 54.16                   | 95.61                 | 93.96                 |
| Meta-Cognitive Editing | <b>MIND(Ours)</b> | 86.12                  | 85.70                   | 79.13                   | 93.43                 | 77.10                 | 75.76                  | 71.82                   | 61.51                   | 90.55                 | 69.54                 |

**Algorithm 1** Algorithmic Overview of the MIND Editing Process

Activation vector  $q_{\text{input}} \in \mathbb{R}^{d'}$ , Target edit label  $y$  Refined activation  $q_{\text{refined}}$

Meta-memory matrix  $\text{Mem} = \{m_i\}_{i=1}^{d'}$

MSV Monitor Network  $\phi(\cdot)$

Prototype bank  $\mathcal{P} = \{\mathbf{m}_k\}_{k=1}^K$

Label projection matrix  $\mathbf{W}_p$ , balance factor  $\beta \in [0, 1]$

► Step 1: Self-Aware Meta-Memory Activation

**for**  $i \leftarrow 1$   $d'$  **do**  $q_{\text{meta}}[i] \leftarrow m_i(q_{\text{input}}[i])$  \*Apply meta-memory transformation

► Step 2: Game-Theoretic Monitoring via MSV  $\phi \leftarrow \text{Softmax}(\text{MSV\_Monitor}(q_{\text{meta}}))$  \*Shapley-based relevance

**for**  $i \leftarrow 1$   $d'$  **do**  $q_r[i] \leftarrow (q_{\text{meta}}[i] \cdot m_i) \cdot \phi[i]$  \*Weighted activation

► Step 3: Reflective Meta-Label Refinement  $p \leftarrow \text{RetrievePrototype}(\mathcal{P}, y)$  \*Get label prototype  $\Delta \leftarrow \mathbf{W}_p \cdot p$  \*Project to label shift space

$q_{\text{refined}} \leftarrow (1 - \beta) \cdot q_r + \beta \cdot \Delta$  \*Final refined activation **return**  $q_{\text{refined}}$

language models without modifying their parameters. IKE constructs context demonstrations to inject new facts, update existing knowledge, and retain unrelated facts.

*WISE*. [53] is designed for lifelong model editing that bridges the gap between long-term and working memory in large language models. *WISE* proposes a dual parametric memory scheme, comprising a main memory for pre-trained knowledge and a side memory for edited knowledge. *WISE* also incorporates a knowledge-sharding mechanism to manage multiple edits efficiently and merge them without conflicts.

### B.3 Implementation Details

For MIND, we adopt the Adam optimizer with a learning rate of  $8e-4$  to update all parameters during the editing stage. Each one-step edit takes approximately 7 seconds to complete. During the pre-training phase of the meta-label refiner, we extract 5K labeled data samples for supervision, and the entire pre-training process takes less than one hour. All experiments are conducted on a computing cluster equipped with five NVIDIA RTX A6000 GPUs.

## C Results on Lifelong Editing Scenario

Lifelong model editing [17, 53] aims to edit the specific model over and over again, which enables large language models to continuously update and adapt their knowledge over time without requiring full retraining.

We also evaluated the effectiveness of our method in a lifelong editing scenario. To ensure fairness and eliminate task-induced biases, experiments were conducted on MMEdit (excluding IKE), as lifelong editing is not meaningful for IKE), with results presented in Table 5. Compared to cognitive-based methods, MIND also demonstrates strong performance. We observe that: (1) Most methods, such as T-Patcher and MEND, exhibit a significant decline in all metrics after multiple editing iterations. However, as  $T$  increases, the rate of decline slows. In contrast, MIND maintains relatively high editing performance even as  $T$  increases. (2) Editing methods specifically designed for lifelong learning, such as *WISE*, achieve high Locality scores. However, they fail to maintain consistently high Reliability and Generality. MIND is able to sustain relatively superior performance compared to other methods, indicating that it can continuously learn meta-knowledge from ongoing edits while maintaining effectiveness.

## D MIND Method Pseudocode Implementation

The pseudocode implementation of the MIND method is presented in Algorithm 1.

## E Ethical Impacts

This work poses no ethical concerns. All experiments are conducted using publicly available datasets and models, without involving any private data or subjective evaluations.